C2.3 Properties of materials

Summary

This section explores the physical properties of elements and compounds and how the nature of their bonding is a factor in their properties.

Underlying knowledge and understanding

Learners will know the difference between an atom, element and compound.

Common misconceptions

Learners commonly have a limited understanding of what can happen during chemical reactions, for example substances may explode, burn, contract, expand or change state.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.
All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
CM2.3i	represent three-dimensional shapes in two dimensions and vice versa when looking at chemical structures, e.g. allotropes of carbon	M5b
CM2.3ii ☑	relate size and scale of atoms to objects in the physical world	M4a
CM2.3iii ☑	estimate size and scale of atoms and nanoparticles	M1d
CM2.3iv ☑	interpret, order and calculate with numbers written in standard form when dealing with nanoparticles	M1b
CM2.3v ☑	use ratios when considering relative sizes and surface area to volume comparisons	M1c
CM2.3vi ☑	calculate surface areas and volumes of cubes	M5c

Topic content		Opportunities to cover:		Practical suggestions
Learning outcomes	**To include**	**Maths**	**Working scientifically**	
C2.3a recall that carbon can form four covalent bonds			WS1.4a	
C2.3b explain that the vast array of natural and synthetic organic compounds occur due to the ability of carbon to form families of similar compounds, chains and rings				
C2.3c explain the properties of diamond, graphite, fullerenes and graphene in terms of their structures and bonding		M5b	WS1.4a	
C2.3d use ideas about energy transfers and the relative strength of chemical bonds and intermolecular forces to explain the different temperatures at which changes of state occur			WS1.2a, WS1.3f, WS1.4a, WS1.4c	
C2.3e use data to predict states of substances under given conditions	data such as temperature and how this may be linked to changes of state			
C2.3f explain how the bulk properties of materials (ionic compounds; simple molecules; giant covalent structures; polymers and metals) are related to the different types of bonds they contain, their bond strengths in relation to intermolecular forces and the ways in which their bonds are arranged	recognition that the atoms themselves do not have the bulk properties of these materials		WS1.4a	

C2.3g ☑	compare ‘nano’ dimensions to typical dimensions of atoms and molecules		M4a, M1d, M1b	WS1.4c, WS1.4d	
C2.3h ☑	describe the surface area to volume relationship for different-sized particles and describe how this affects properties		M1c	WS1.4c	Dissolving tablets. (PAG C8)
C2.3i ☑	describe how the properties of nanoparticulate materials are related to their uses		M5c	WS1.1c, WS1.1e, WS1.3c, WS1.4a	
C2.3j ☑	explain the possible risks associated with some nanoparticulate materials			WS1.1d, WS1.1f, WS1.1h, WS1.1i WS1.4a	